

Bike Rental Demand Forecasting

UE24CS352A · Machine Learning Mini-Project · Batch 19

PES2UG24CS063 · Aniket Raosaheb Jadhav | PES2UG24CS109 · Basava Prabhu

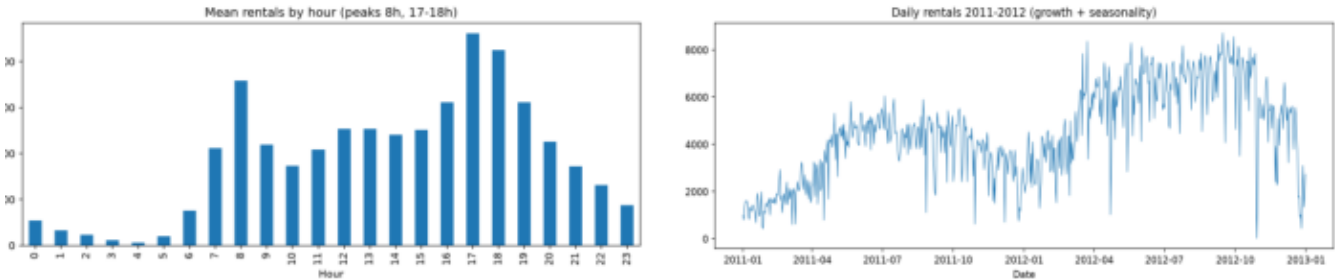
1 Problem Statement

Bike-sharing services need to anticipate how many bikes will be required at different times. We used two years of Washington, D.C. rental records (2011–2012), together with time and weather information, to predict the total number of rentals. The data was split chronologically so that training uses the past and testing represents the future. Four approaches were compared.

2 Dataset & Exploratory Findings

hour.csv contains 17,379 hourly records and **day.csv** contains 731 daily records. Features include date/time information, season, month, weekday, holiday, working-day status, weather type, temperature, feels-like temperature, humidity and wind speed. The prediction target is **count (cnt)**, the total of casual and registered rentals. The report found no missing values or duplicates.

Demand shows both seasonal growth and strong intraday structure. Working days show pronounced morning and evening peaks, while the daily series shows seasonal waves and year-over-year growth.



Hourly demand shows clear morning/evening peaks; daily demand shows growth together with seasonal variation.

3 Methodology

Date information was transformed into hour, weekday, month, weekend and rush-hour features. Circular encodings were added for time-of-day continuity. Temperature and feels-like temperature were highly correlated, so only feels-like temperature was retained. Season and weather type were treated as categorical variables. Scaling and encoding were fitted on past data only and then applied to future data to avoid leakage.

Four models were evaluated: an average-demand baseline, Linear Regression, Random Forest and Gradient Boosting. The evaluation used chronological testing with error measures that capture both average absolute error and larger mistakes, together with RMSLE for relative/log-scale performance.

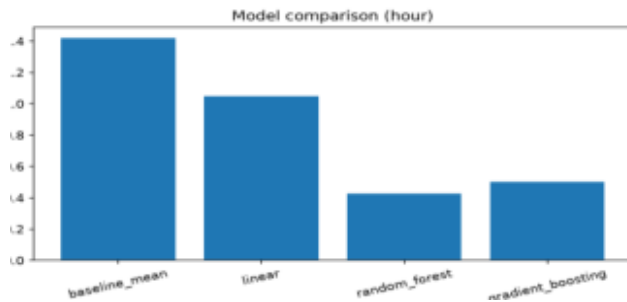
4 Results

Model	MAE (bikes)	RMSE	RMSLE
Gradient Boosting	52.6	77.3	0.50
Random Forest	57.1	84.8	0.43 (best)
Linear Regression	94.3	126.7	1.05
Average baseline	176.8	237.7	1.42

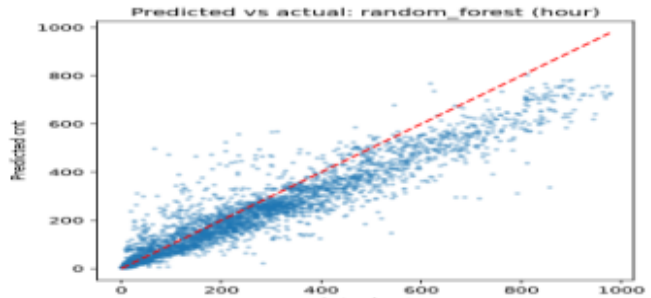
Interpretation: Gradient Boosting has the lowest MAE and RMSE, while Random Forest has the lowest RMSLE. Both tree-based models substantially outperform the average baseline, showing that demand is driven by nonlinear interactions between time, commuting patterns and weather.

Bike Rental Demand Forecasting — Analysis & Conclusions

Model behaviour, feature importance, error analysis and implementation

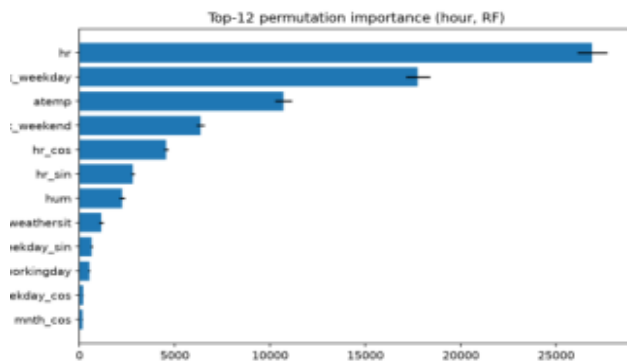


Left: RMSLE comparison, where lower is better. Right: Random Forest predictions versus actual hourly demand.

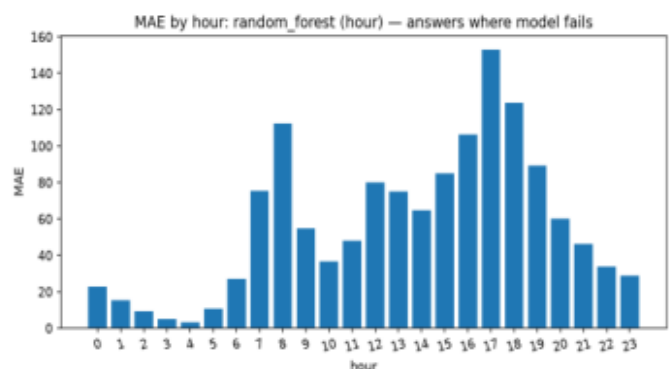


5 What the Model Learns

The hour of the day is the strongest signal, followed by rush-hour features and feels-like temperature. Removing groups of rush-hour, time-cycle or weather features reduces performance. This supports the idea that demand is strongly structured by commuting schedules and environmental conditions.



Feature importance highlights time and rush-hour variables; hourly error is largest around the evening peak.



6 Error Analysis & Limitations

Errors are highest around the 5pm peak and during bad-weather periods. High-demand hours naturally produce larger absolute errors. The daily dataset is also relatively small, and rare bad-weather conditions have limited examples, which restricts how confidently those cases can be modeled.

7 Implementation & Contribution

The project is organized into **src/** for cleaning, training, evaluation and prediction, **notebooks/01_edu.py** for analysis, and **reports/figures** for generated figures. Aniket handled data cleaning and exploration; Basava handled training and validation; evaluation, the report and the demo were joint. The project can be reproduced using the requirements file and the execution order documented in the README.

8 Conclusion & Future Work

The results show that a relatively compact feature set combined with a tree-based model can forecast hourly bike demand substantially better than a simple average or a linear model. The main takeaway is that temporal structure—especially hour and rush-hour behaviour—is essential for this task. Future work would focus on hyperparameter tuning and a deeper investigation of peak-hour errors.

Overall conclusion: tree-based models capture the nonlinear demand patterns more effectively, while chronological evaluation provides a realistic view of forecasting performance.